

Benchmarking Class-Imbalance Mitigation Across Four Histopathology Datasets

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Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report benchmarks representative class-imbalance mitigation methods on four histopathology datasets—CAMELYON16, TCGA-UT, BRACS, and PANDA—presented in order of the diagnostic question they pose (detection, typing, subtyping, grading) and chosen to span the difficulty–imbalance plane while varying organ, label type, and annotation form. Imbalance is characterised throughout by the entropy-based measure $1 - H_{\text{norm}} \in [0, 1]$, reported separately at patch and slide level. All benchmarks fix a frozen Virchow2 encoder, a bounded per-slide evidence budget, and patient-disjoint splits, so that comparisons within each regime isolate the imbalance mechanism. The four datasets place the same methods at very different points of the plane. On CAMELYON16 (severe patch imbalance $1 - H_{\text{norm}} \approx 0.83$, near-balanced slide labels ≈ 0.03 , separable detection task) every method reaches near-ceiling discrimination and the unweighted baseline is competitive. On TCGA-UT (mild imbalance ≈ 0.09 , 32-class typing) centre-focused affinity loss (CFAL) reaches the highest patch-level macro F1, OKO set learning the highest balanced accuracy and the best raw NLL and Brier among patch methods, while RankMix and MDE-MIL lead the WSI-bag regime. On BRACS (mild patch imbalance ≈ 0.07 , hard fine-grained subtyping) only OKO and CFAL separate from the CE baseline. On PANDA (mild imbalance, two targets of very different difficulty on the same biopsies) the no-mitigation baseline is competitive in both regimes. Together the four datasets show that the headroom for imbalance mitigation is governed by class separability in the frozen representation, not by the imbalance of the class distribution.

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1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive metrics. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Imbalance methods often assume a specific prediction unit. Image-level methods use labeled images, image embeddings, or directly generated image samples, whereas WSI-level methods use weak labels over bags of instances together with attention, instance ranking, or bag-level representation learning. The computational-pathology pipeline these regimes sit in—whole-slide imaging, tiling into patches, and weakly supervised multiple-instance learning over frozen foundation-model features—and the full method definitions, imbalance mechanisms, and dataset provenance are documented in two companion reports (Queisler, 2026a,b); this report contributes the experimental results and cites them only where a definition is needed.

The benchmark asks one question across four datasets, presented in order of the diagnostic question they pose—CAMELYON16 detection, TCGA-UT typing, BRACS subtyping, PANDA grading: when the encoder is a strong frozen foundation model, which imbalance mechanism helps in each input regime, and what property of the data governs whether any mitigation helps at all? The four datasets span the difficulty–imbalance plane by design. CAMELYON16 is severely imbalanced at patch level but separable; TCGA-UT is a mildly imbalanced many-class typing anchor; BRACS is mildly imbalanced but hard; and PANDA supplies two mildly imbalanced targets of different difficulty on the same biopsies. This spread separates the roles of imbalance severity and class separability: if the imbalance of the class distribution governed mitigation headroom, CAMELYON16 should show the largest gains; if class separability governs it, CAMELYON16 should show the smallest gains despite being the most imbalanced.

2 Datasets

The four datasets are summarised in Table 1; this section states only what the benchmark consumes from each. Imbalance is quantified by an entropy-based measure. For empirical class proportions $p_1, \dots, p_K$ (with $\sum_i p_i = 1$), the Shannon entropy $H(p) = -\sum_{i=1}^K p_i \log p_i$ attains its maximum $\log K$ at the uniform distribution; normalising and complementing gives

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{i=1}^K p_i \log p_i}{\log K} \in [0, 1], \quad (1)$$

where 0 is a perfectly balanced distribution and values closer to 1 indicate stronger imbalance. Because the $\log K$ normalisation removes the dependence on the number of classes, the measure compares a two-class cohort against a thirty-two-class one fairly, unlike the raw head-to-tail (largest-to-smallest class support) ratio. It is reported separately at patch and slide level, because the two can differ sharply when patch labels are derived from masks rather than inherited from the slide; the full native class distributions are given in Queisler (2026a). All benchmarks use frozen Virchow2 2,560-dimensional features (concatenation of CLS token and mean patch token, cast to float32) and patient-disjoint train/validation/test splits across three random seeds (Vorontsov et al., 2024; Zimmermann et al., 2024).

CAMELYON16. Binary labels are mask-derived: each tile is labelled tumor or normal by intersecting it with the published annotation mask. Of 399 slides, 398 have usable tile evidence; the 20 tumor slides whose annotations are not exhaustive are excluded from the patch regime but retained in the WSI-bag regime, where the reliable slide-level label is used. The per-slide

Dataset	Target label(s)	Split unit	Budget (med. tiles)	Imbalance $1 - H_{\text{norm}}$	
				Patch	Slide
CAMELYON16	Tumor/normal (patch); metastasis (WSI)	Slide = patient	7874	0.832	0.028
TCGA-UT	32 cancer types	Case (patient)	30	0.085	0.085
BRACS	7 breast-lesion subtypes	Patient	360	0.072	0.018
PANDA	Cancer/benign (patch); ISUP 0–5 (WSI)	Biopsy	421	0.332	0.043

Table 1: Benchmark dataset overview, in the order the datasets are discussed. The evidence budget is the per-slide cap (data-driven median of available tiles) used in both regimes. Imbalance is the entropy-based $1 - H_{\text{norm}}$; for PANDA the patch and slide figures describe two different targets (binary cancer/benign vs. six-class ISUP).

evidence budget is the empirical median of available tissue tiles (7,874)—two orders of magnitude larger than TCGA-UT’s 30-tile cap, reflecting that a whole lymph-node section yields far more tissue than a single lesion ROI or tumor-region crop. Each slide is its own patient, so slide-disjoint and patient-disjoint splits coincide. The severe patch imbalance ($1 - H_{\text{norm}} \approx 0.832$) is a within-slide effect: tumor tiles are scarce even on slides that do contain a metastasis, so the slide-level labels are nearly balanced (≈ 0.028).

TCGA-UT. The dataset contains 1,608,060 patches from 8,736 diagnostic slides across 32 cancer types (Tomczak et al., 2015; Komura et al., 2022). Splits are assigned at the TCGA participant level (case-disjoint), using 5,025 training, 1,075 validation, and 1,075 test cases per seed; all slides from one participant remain in one partition. Because every patch inherits its slide’s cancer-type label, the patch and slide imbalance coincide ($1 - H_{\text{norm}} \approx 0.085$). The patch benchmark caps each slide at 30 deterministic patches (empirical median of available tiles at resolution level 0), yielding a controlled manifest shared by all patch methods; the WSI-bag benchmark applies the same 30-instance cap to each slide’s feature bag.

BRACS. Evidence is derived from the ROI set: each annotated ROI is tiled into 256×256 RGB patches, and each slide’s pooled ROI tiles are capped at the empirical median (360 tiles per slide). This design preserves label fidelity because all instances originate from pathologist-selected lesion regions. Of 547 published WSIs, 387 contribute at least one ROI tile; the benchmark covers 151 patients and 99,352 extracted patches over the seven-subtype target, a mild imbalance ($1 - H_{\text{norm}} \approx 0.072$ patch, ≈ 0.018 slide).

PANDA. The WSI-bag regime uses the canonical six-class ISUP grade (0 = benign, 1–5 = increasing cancer grade groups; $1 - H_{\text{norm}} \approx 0.043$). The patch regime uses a provider-consistent binary cancer/benign target derived from pixel masks ($1 - H_{\text{norm}} \approx 0.332$); because Radboud and Karolinska masks use institution-dependent value semantics, collapsing both to cancer/benign is the only label the two providers share. To keep frozen-feature extraction tractable, the benchmark uses a stratified subset of approximately 2,000 biopsies, sampled to preserve the joint distribution of ISUP grade and provider. The per-slide evidence budget is the empirical median number of tissue tiles per biopsy (421); biopsies without available masks are excluded from the patch regime but kept in the WSI-bag regime.

Native imbalance is mild at the level the benchmark evaluates. Table 1 makes a second point beyond the per-dataset summaries above: at *slide* level—the unit every diagnostic target in this benchmark is ultimately scored on—all four datasets are only mildly skewed ($1 - H_{\text{norm}} \in [0.018, 0.085]$). The two large values in the table (CAMELYON16 patch 0.832, PANDA

patch 0.332) are within-slide, binary mask effects, not properties of the typing, subtyping, or grading targets themselves. Figure 1 places these eight numbers against the imbalance factors long-tailed CV benchmarks use to stress-test the very mitigation methods evaluated here: Cui et al. construct long-tailed CIFAR-10/100 by an exponential class-count decay at imbalance factor $\rho = N_{\max}/N_{\min} \in \{10, \dots, 200\}$ (Cui et al., 2019), and Buda et al. study step and linear imbalance at comparable ratios (Buda et al., 2018); at $\rho = 100$ this places $1 - H_{\text{norm}}$ at 0.29 for CIFAR-10-LT and 0.13 for CIFAR-100-LT (shaded band). Every native slide-level target sits below this band, and three of the four sit below even the more conservative CIFAR-100-LT end. Benchmarking imbalance-mitigation methods on these native distributions therefore under-tests them: the methods were designed for, and validated on, a regime of imbalance the native slide-level targets do not reach. §3.1 proposes a controlled-imbalance construction that closes this gap for a future benchmark iteration.

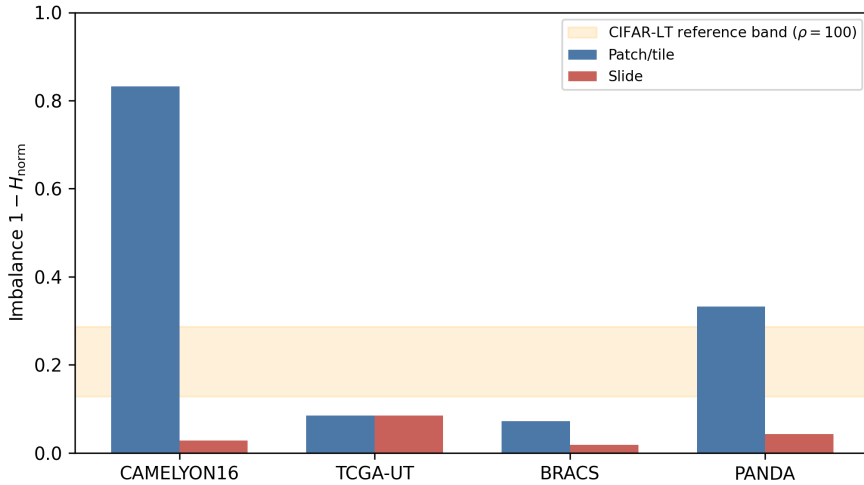


Figure 1: Native $1 - H_{\text{norm}}$ at patch/tile and slide level for all four datasets, against a reference band for CIFAR-10-LT/CIFAR-100-LT at imbalance factor $\rho = 100$ (Cui et al., 2019). Every slide-level target—the unit each diagnostic question is scored on—sits below the band.

All four datasets use three random seeds, deterministic per-slide evidence caps, and frozen Virchow2 features. Hyperparameters are selected independently per method: for each candidate configuration the validation macro F1 is averaged over the three seeds and the configuration with the highest mean is chosen (balanced accuracy as tie-breaker); a single configuration per method is then fixed before evaluating on the held-out test split. Results are reported as mean $\pm$ standard deviation across seeds, and patch and WSI-bag methods are compared only within their own regime.

3 Experimental Setup

The methods, training protocol, and tuning grids below are shared across all four datasets; full method definitions and metric formulas follow Queisler (2026b). Table 2 lists the method roster by family. CE and MIL CE are unweighted cross-entropy baselines, included so that mitigation gains are measured against a genuine no-mitigation reference; patch and WSI-bag methods are never ranked against each other.

Both benchmarks share a fixed training protocol (Table 3): AdamW at learning rate 10^{-3} , weight decay 10^{-4} , 30 epochs with no early stopping, final-epoch model selection, and seeds 0–2. The only per-method degree of freedom is a single imbalance-specific control, swept on

Family	Methods	Regime
No-mitigation baseline	CE, MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling-loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 2: Method families and regimes. Method definitions follow Queisler (2026b).

validation using the grids in Table 4; identical grids are applied on all four datasets. ProGAN sweeps only the final-depth refinement epochs, with generator training following Table 5.

Setting	Patch benchmark	WSI-bag benchmark
Optimizer	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}
Epochs	30	30
Early stopping	none	none
Minibatch size	256 patches	32 slide bags
Model selection	final epoch	final epoch
Random seeds	0, 1, 2	0, 1, 2

Table 3: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted in Queisler (2026b).

The patch and WSI-bag classifier heads (Figure 2) are shared across all datasets. The patch head is a two-layer MLP (linear map to 512 units, ReLU, dropout 0.1, linear output to C classes); CFAL keeps the trunk but replaces the final linear map with one learnable prototype per class and Gaussian affinity-based prediction. The WSI-bag head is an attention-based MIL network: each instance passes through a shared encoder ($2560 \rightarrow 256$), instance weights come from a linear scorer with softmax over instances, the bag representation is their weighted sum, and slide logits come from a final linear map. SC-MIL adds a two-layer projection head of width 256; MDE-MIL keeps the attention pooling but uses two expert heads. The WSI-bag input is capped at the dataset-specific evidence budget (Table 1).

Macro F1 is the primary ranking metric within each regime because it weights classes equally and penalizes failure on rare classes; balanced accuracy and accuracy are reported alongside it. Three complementary probability-quality metrics—negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE)—are reported on held-out softmax outputs, both before and after a unified post-hoc temperature-scaling protocol in which, for each method and seed, a single temperature is fit on validation by minimizing NLL and applied unchanged to the test scores. Full metric definitions and the temperature-scaling protocol are given in Queisler (2026b).

3.1 Controlled imbalance construction

Section 2 showed that every slide-level target is only mildly imbalanced relative to the CIFAR-LT regime the evaluated mitigation methods were designed for. Rather than accepting that mismatch, each dataset can be restructured to a controlled, artificially skewed distribution

Regime	Method	Reference default	Validation grid
Patch	Weighted CE	Weight power = 1	{0.25, 0.5, 0.75, 1}
WSI bag	Weighted CE	Weight power = 1	{0, 0.125, 0.25, 0.5, 0.75, 1}
Patch, WSI	Balanced sampler	Sampler power = 1	{0.5, 0.75, 1}
Patch, WSI	Focal loss	$\gamma = 2$	$\gamma \in \{0.5, 1, 1.5, 2\}$
Patch	CE + soft F1	Metric-loss weight = 1	{0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 256, 512, 1024}
Patch	CE + soft MCC	Metric-loss weight = 1	{0.25, 0.5, 1, 2, 8, 32}
Patch	CFAL	$\gamma = 2, \sigma = 1$	$\sigma \in \{0.25, 1, 4\}$
Patch	ProGAN augmentation	Final-depth epochs = 25	{10, 25, 50}
Patch	OKO	$k = 1$	$k \in \{1, 2, 3, 4, 5, 6, 8, 12, 20\}$
WSI bag	RankMix	$\alpha = 1$	$\alpha \in \{0.2, 0.5, 1, 2, 4, 8, 16, 32, 128\}$
WSI bag	SC-MIL	$\tau = 1$	$\tau \in \{0.05, 0.1, 0.2, 0.5\}$
WSI bag	MDE-MIL	$\lambda_{\text{con}} = 0.25$	$\lambda_{\text{con}} \in \{0.1, 0.25, 0.3, 0.5, 1, 2, 4\}$

Table 4: Method-specific imbalance controls. Each row varies one control on validation while retaining the shared budget in Table 3. Identical grids are applied on all four datasets.

Setting	Value
Resolution stages	4×4 to 256×256 (seven stages)
Epochs per stage (depths 1–6)	25
Final-depth (256×256) refinement epochs	25 (reference); validation selects from {10, 25, 50}
Fade-in fraction for new layers	0.5 (pinned to the reference 25-epoch schedule)
Latent dimension	128
Base channel width	512
Adversarial loss	WGAN-GP ($\lambda = 10$, drift = 10^{-3})
Equalized learning rate	Yes (weight $\sim \mathcal{N}(0, 1)$, scale $\sqrt{2/\text{fan_in}}$ at forward)
Optimizer	Adam, learning rate 2×10^{-4} , $\beta_1 = 0.5$
Real patches per class for GAN training	Up to 2,048, deterministic per-class subsample
Synthetic count per tail class	Head-class train count minus class train count

Table 5: ProGAN generator training settings (TCGA-UT patch regime only). Depths 1–6 use 25 epochs each; for the final depth one long run trains to 50 epochs, snapshotting the generator at epochs 10, 25, and 50, and validation selects the snapshot with the best mean macro F1 over seeds, reusing one GAN training run for all three sweep points.

using two constructions from the long-tailed-recognition literature, both of which enforce a hard minimum support N_{floor} so that no class is starved below what is needed for stable evaluation.

Truncated exponential decay (Cui et al.). Ranking classes by native support, the target count for the i -th ranked class ($i = 0, \dots, K - 1$) is

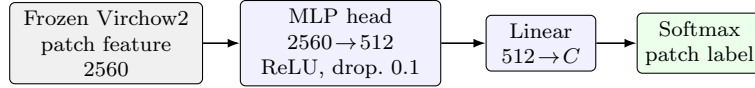
$$n_i = \max(N_{\text{floor}}, \lfloor N_{\text{max}} \mu^i \rfloor), \quad \mu \in (0, 1), \quad (2)$$

with μ chosen to hit a target imbalance factor $\rho = N_{\text{max}}/N_{\text{min}}$ (Cui et al., 2019). Unlike the original CIFAR-LT construction, which lets the tail decay to as few as a handful of images, N_{floor} caps the decay so every class retains enough slides for macro-averaged evaluation.

Step imbalance (Buda et al.). Classes are split into a head group, which keeps its native support, and a tail group, which is uniformly downsampled to N_{floor} (Buda et al., 2018). This isolates rarity as a binary property (in the head group or not) rather than a continuous rank effect, and lets the tail group be chosen for clinical relevance rather than by native rank.

Per-dataset strategy. Table 6 applies these constructions, plus simple ratio downsampling for the binary CAMELYON16 case, to each dataset’s native slide counts:

(a) Patch benchmark



(b) WSI-bag benchmark

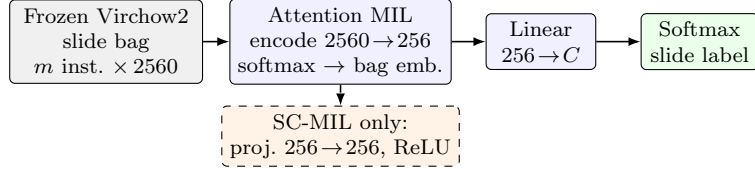


Figure 2: Shared classifier heads for the benchmark (C = number of classes in each dataset). Panel (a) classifies one controlled patch embedding per forward pass; CFAL reuses the MLP trunk but replaces the linear $512 \rightarrow C$ head with learnable prototypes and affinity-based prediction. Panel (b) encodes all instances on a slide, pools them with attention, and predicts one slide-level label; the dashed branch is used only by SC-MIL, and MDE-MIL replaces the single linear head with two expert heads. Virchow2 features are frozen in both benchmarks.

- **TCGA-UT** (32 classes) uses exponential decay ranked by native support, floored at $N_{\text{floor}} = 20$, so rare types such as DLBC and cholangiocarcinoma remain evaluable while the head-to-tail ratio widens well past its native 28.3:1.
- **PANDA** (6 ordinal ISUP grades) uses the same decay but fixed to the ordinal grade order rather than re-ranked by support, so the artificial skew still respects grade order; the aggressive ISUP 4–5 grades are floored at $N_{\text{floor}} = 50$.
- **BRACS** (7 subtypes) uses step imbalance: the benign/usual head group {N, PB, UDH, IC} keeps native support, while the clinically critical atypical/malignant tail {FEA, ADH, DCIS} is floored at $N_{\text{floor}} = 30$.
- **CAMELYON16** (binary) is downsampled to a fixed 10:1 normal:tumor ratio, floored at $N_{\text{floor}} = 20$ tumor slides, simulating a low-prevalence screening scenario.

Figure 3 shows the resulting native-versus-target support for all four datasets; Table 6 reports the resulting head-to-tail ratio and $1 - H_{\text{norm}}$. This construction is proposed as the basis for a future benchmark iteration; the results in §4 still use the native distributions.

Dataset	Method	K	N_{floor}	ρ (native $\rightarrow$ target)	$1 - H_{\text{norm}}$ (native $\rightarrow$ target)
TCGA-UT	Exp. decay	32	20	28.3:1 $\rightarrow$ 39.6:1	0.085 $\rightarrow$ 0.126
BRACS	Step	7	30	2.3:1 $\rightarrow$ 6.5:1	0.018 $\rightarrow$ 0.113
PANDA	Ordinal exp. decay	6	50	2.4:1 $\rightarrow$ 10.9:1	0.043 $\rightarrow$ 0.153
CAMELYON16	Ratio downsampling (10:1)	2	20	1.5:1 $\rightarrow$ 9.9:1	0.028 $\rightarrow$ 0.558

Table 6: Per-dataset controlled-imbalance construction: method, class count K , floor N_{floor} , and the resulting head-to-tail ratio and entropy-based imbalance, native versus target.

4 Results

The four datasets are reported in the order of Table 1. Each dataset carries one results float stacking the patch-feature and WSI-bag regimes, followed by three short paragraphs on patch-

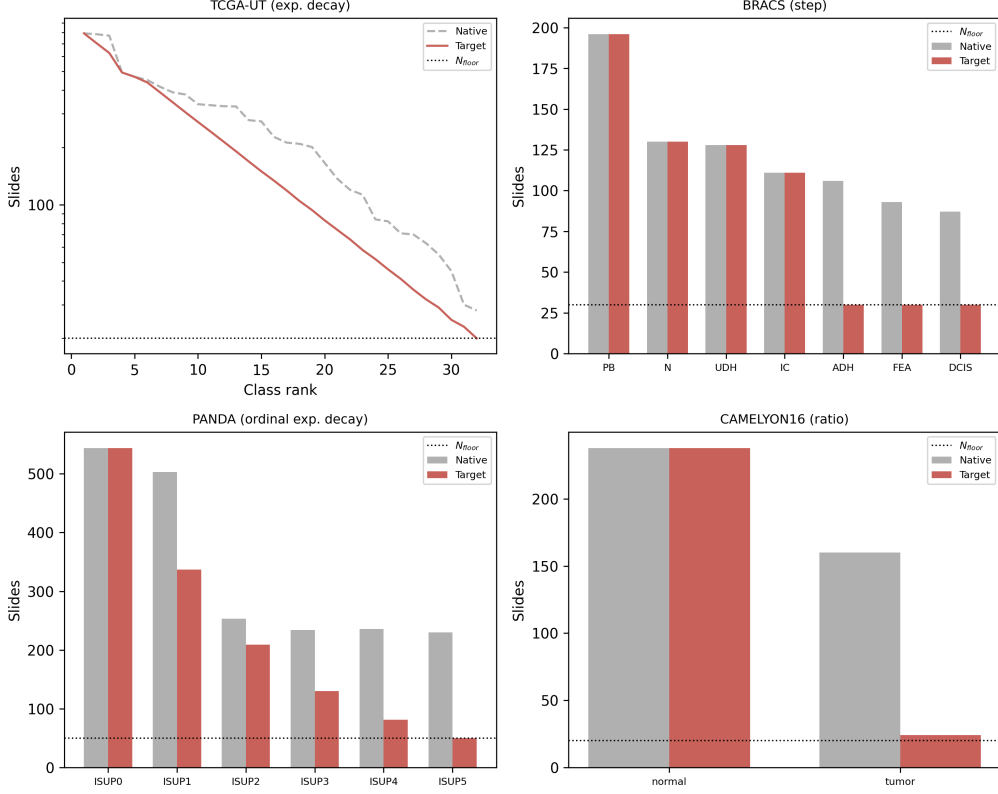


Figure 3: Native (grey) versus proposed controlled-imbalance target (red) slide support per dataset, with the hard floor N_{floor} (dotted). TCGA-UT and PANDA use truncated exponential decay (Equation 2), ranked by support and by ordinal grade respectively; BRACS uses step imbalance; CAMELYON16 uses ratio downsampling.

level results, bag-level results, and calibration. Absolute scores and method orderings are never comparable across the two regimes. TCGA-UT receives the deepest analysis; its tuning-sensitivity, paired-seed, and classwise-difficulty diagnostics, and all raw calibration tables, are collected in Appendices A and B.

4.1 CAMELYON16 — metastasis detection

CAMELYON16 is the most imbalanced dataset at patch level ($1 - H_{\text{norm}} \approx 0.832$) yet nearly balanced at slide level (≈ 0.028), and its detection task is the most separable in the suite. The same frozen features separate tumor from normal tissue almost perfectly, so despite the severe patch imbalance every method reaches near-ceiling discrimination in both regimes (Table 7).

In the patch regime, balanced accuracy ranges from 0.956 to 0.986 and the unweighted CE baseline (BAcc 0.962) sits squarely inside the field. In the WSI-bag regime, all methods score between balanced accuracy 0.965 and 0.984; the unweighted MIL CE baseline is again fully competitive (BAcc 0.965), and the entire field is separated by less than 0.02 macro F1—canonical slide-level metastasis detection is nearly solved from bounded Virchow2 feature bags. Calibration follows discrimination (Appendix B): raw patch ECE is 0.003–0.005 for every method except CFAL, and temperature scaling brings it to 0.001–0.002, markedly better than BRACS—the calibration counterpart of the separability gap.

Patch-feature regime

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.997 ± 0.002	0.961 ± 0.015	0.966 ± 0.008
Balanced sampling	0.997 ± 0.001	0.969 ± 0.014	0.960 ± 0.020
ProGAN augmentation	0.996 ± 0.001	0.956 ± 0.009	0.959 ± 0.019
CE	0.996 ± 0.001	0.962 ± 0.015	0.958 ± 0.020
Focal loss	0.996 ± 0.001	0.972 ± 0.010	0.957 ± 0.027
OKO	0.996 ± 0.000	0.974 ± 0.007	0.955 ± 0.034
CE + soft F1	0.996 ± 0.001	0.986 ± 0.002	0.952 ± 0.035
Weighted CE	0.996 ± 0.002	0.966 ± 0.012	0.952 ± 0.028
CE + soft MCC	0.995 ± 0.001	0.982 ± 0.006	0.940 ± 0.050

WSI-bag regime (slide-level metastasis detection)

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
SC-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
Balanced sampling	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
Focal loss	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
RankMix	0.972 ± 0.020	0.974 ± 0.020	0.971 ± 0.020
MIL CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018
Weighted CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018

Table 7: CAMELYON16 native test results for validation-selected configurations, mean $\pm$ standard deviation across seeds.

4.2 TCGA-UT — pan-cancer typing

TCGA-UT widens the task to 32-class pan-cancer typing at a mild imbalance ($1 - H_{\text{norm}} \approx 0.085$, identical at patch and slide level because every patch inherits its slide label). It receives the study’s deepest analysis: how much the selected control matters varies sharply across methods—across-configuration variation exceeds twice the seed standard deviation only for OKO and RankMix, while every single-knob resampling, reweighting, and focal mechanism produces variation within seed noise (Appendix A).

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings. CFAL ranks highest on macro F1 and OKO highest on balanced accuracy, both with paired gains over plain CE in all three seeds (Appendix A); weighted CE, balanced sampling, and focal loss remain within the CE-centered band. ProGAN augmentation stays within the non-generative spread despite adding a large synthetic tail-class volume, with per-class Inception FID values comparable to a source study where GAN augmentation helped an end-to-end classifier (Ruiz-Casado et al., 2024); the frozen encoder—not generator quality—is the proposed explanation for the null downstream effect (Appendix A).

On WSI bags, RankMix has the highest mean balanced accuracy and MDE-MIL the highest mean macro F1, both above plain MIL CE on both headline metrics in every seed. Other MIL variants cluster closely below this pair, and focal MIL is unstable across seeds (competitive on seeds 0 and 2, collapsing on seed 1), giving the widest seed-to-seed standard deviation in the WSI benchmark. On raw probability quality, CFAL improves discrimination most but is the most over-confident patch method (highest raw ECE, worst NLL/Brier), while OKO is the

<i>Patch-feature regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.847 ± 0.008	0.789 ± 0.007	0.798 ± 0.009
OKO	0.841 ± 0.006	0.791 ± 0.005	0.788 ± 0.003
CE + soft F1 (balanced)	0.835 ± 0.004	0.781 ± 0.002	0.780 ± 0.003
Weighted CE	0.826 ± 0.008	0.767 ± 0.006	0.774 ± 0.005
CE	0.828 ± 0.010	0.764 ± 0.011	0.773 ± 0.015
Balanced sampler	0.827 ± 0.010	0.769 ± 0.007	0.771 ± 0.014
CE + soft MCC (balanced)	0.824 ± 0.009	0.772 ± 0.004	0.770 ± 0.005
ProGAN augmentation	0.824 ± 0.004	0.760 ± 0.005	0.766 ± 0.002
Focal	0.825 ± 0.007	0.758 ± 0.007	0.765 ± 0.009

<i>WSI-bag regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL (ensemble)	0.896 ± 0.002	0.849 ± 0.014	0.865 ± 0.011
RankMix	0.890 ± 0.005	0.864 ± 0.002	0.854 ± 0.008
SC-MIL	0.883 ± 0.013	0.837 ± 0.013	0.839 ± 0.008
Weighted MIL	0.882 ± 0.006	0.833 ± 0.019	0.835 ± 0.011
Balanced MIL	0.875 ± 0.003	0.829 ± 0.010	0.831 ± 0.009
MIL CE	0.873 ± 0.013	0.823 ± 0.014	0.826 ± 0.013
Focal MIL	0.872 ± 0.006	0.823 ± 0.019	0.824 ± 0.008

Table 8: TCGA-UT validation-selected test results over three seeds. Methods share the frozen Virchow2 encoder, the shared head family, and the evaluation code within each regime.

exception to the usual discrimination–calibration trade-off, attaining the best raw NLL and Brier among patch methods; focal loss attains the smallest raw ECE. On WSI bags, MDE-MIL and RankMix reach the lowest raw NLL, Brier, and ECE. After temperature scaling, CFAL reaches the best patch NLL and Brier and focal loss the lowest patch ECE, while the simpler MIL baselines reach lower post-hoc ECE than MDE-MIL or RankMix, so the raw WSI ECE advantage should be read partly as a confidence-scale mismatch (Appendix B).

4.3 BRACS — breast-lesion subtyping

BRACS returns to a single organ but sharpens to fine-grained seven-subtype breast-lesion classification at a mild imbalance ($1 - H_{\text{norm}} \approx 0.072$ patch, ≈ 0.018 slide). Absolute scores are far lower than on TCGA-UT because the task is intrinsically harder and the breast-only training pool is smaller; even task-specific BRACS models report only mid-60% weighted F1 (Pati et al., 2022; Stegmüller et al., 2023), so a frozen-feature benchmark under a bounded budget should not approach TCGA-UT-level discrimination.

In the patch regime, OKO and CFAL are the only methods that visibly separate from the unweighted CE baseline, leading by roughly five balanced-accuracy points, while every other method—including ProGAN augmentation—falls inside one standard deviation of CE. In the WSI-bag regime, the unweighted MIL CE baseline has the highest balanced accuracy and macro F1 and no mitigation method improves on it—exactly the failure mode the no-mitigation baseline is included to expose. Calibration is severe: raw patch ECE is 0.29–0.42 for most methods, with OKO the exception (raw ECE 0.045, fitted temperature $T \approx 0.88$) and CFAL showing the near-degenerate low-temperature signature ($T \approx 0.08$) that scaling nonetheless corrects; WSI-bag calibration is uniformly worse (post-scaling ECE 0.136–0.168), reflecting the smaller training pool and fine-grained label set (Appendix B).

<i>Patch-feature regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.589 ± 0.027	0.497 ± 0.042	0.490 ± 0.038
OKO	0.565 ± 0.054	0.502 ± 0.052	0.488 ± 0.049
CE + soft F1	0.547 ± 0.034	0.480 ± 0.036	0.468 ± 0.033
CE + soft MCC	0.545 ± 0.043	0.472 ± 0.047	0.463 ± 0.044
Balanced sampling	0.552 ± 0.045	0.470 ± 0.043	0.461 ± 0.045
ProGAN augmentation	0.549 ± 0.049	0.460 ± 0.045	0.457 ± 0.049
Focal loss	0.533 ± 0.041	0.460 ± 0.037	0.452 ± 0.036
CE	0.548 ± 0.045	0.451 ± 0.056	0.451 ± 0.055
Weighted CE	0.531 ± 0.045	0.461 ± 0.040	0.450 ± 0.042
<i>WSI-bag regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
MIL CE	0.672 ± 0.020	0.524 ± 0.103	0.508 ± 0.045
Weighted CE	0.667 ± 0.029	0.510 ± 0.093	0.501 ± 0.055
Focal loss	0.691 ± 0.029	0.490 ± 0.032	0.498 ± 0.033
MDE-MIL	0.635 ± 0.033	0.489 ± 0.093	0.480 ± 0.051
Balanced sampling	0.635 ± 0.022	0.449 ± 0.022	0.462 ± 0.023
RankMix	0.603 ± 0.104	0.437 ± 0.087	0.446 ± 0.075
SC-MIL	0.539 ± 0.092	0.367 ± 0.083	0.383 ± 0.086

Table 9: BRACS native test results for validation-selected configurations, mean $\pm$ standard deviation across patient-disjoint seeds.

4.4 PANDA — prostate cancer grading

PANDA poses two mildly imbalanced targets on the same biopsies at opposite ends of the difficulty axis: binary cancer/benign tile detection ($1 - H_{\text{norm}} \approx 0.332$, patch regime) and six-class ordinal ISUP grading (≈ 0.043 , WSI-bag regime), the latter a fine-grained ordinal problem on which even purpose-built challenge models reach only moderate agreement (Bulten et al., 2022).

In the patch regime, every method reaches balanced accuracy between 0.832 and 0.860 and macro F1 between 0.811 and 0.866; the unweighted CE baseline sits squarely inside this band (BAcc 0.838, macro F1 0.859) and no mitigation method separates from it by more than between-seed noise. The WSI-bag regime inverts the difficulty: macro F1 spans only 0.559–0.590—roughly 0.27 below the detection regime on the very same biopsies—and the unweighted MIL CE baseline (BAcc 0.566, macro F1 0.565) is within noise of the field; balanced sampling, focal loss, and MDE-MIL lead it by about 0.02, but between-seed standard deviations of 0.02–0.04 make these numerical rather than statistically separated leads. Calibration follows separability: patch raw ECE is 0.024–0.103 and post-scaling ECE 0.005–0.036 (OKO best), whereas the harder WSI-bag regime has raw ECE 0.129–0.252 and residual post-scaling ECE 0.039–0.097 (Appendix B).

Patch-feature regime (cancer vs. benign tiles)

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.928 ± 0.004	0.832 ± 0.003	0.866 ± 0.002
CE	0.921 ± 0.004	0.838 ± 0.003	0.859 ± 0.001
ProGAN augmentation	0.918 ± 0.005	0.839 ± 0.000	0.855 ± 0.002
Weighted CE	0.915 ± 0.006	0.845 ± 0.004	0.854 ± 0.003
Focal loss	0.896 ± 0.014	0.854 ± 0.006	0.834 ± 0.012
OKO	0.892 ± 0.004	0.860 ± 0.006	0.832 ± 0.006
Balanced sampling	0.890 ± 0.004	0.848 ± 0.003	0.826 ± 0.002
CE + soft F1	0.876 ± 0.009	0.852 ± 0.006	0.813 ± 0.005
CE + soft MCC	0.874 ± 0.009	0.852 ± 0.006	0.811 ± 0.004

WSI-bag regime (six-class ISUP grading)

Method	Accuracy	Balanced accuracy	Macro F1
Balanced sampling	0.653 ± 0.028	0.588 ± 0.032	0.590 ± 0.033
Focal loss	0.654 ± 0.003	0.584 ± 0.016	0.583 ± 0.018
MDE-MIL	0.652 ± 0.029	0.581 ± 0.036	0.581 ± 0.040
RankMix	0.626 ± 0.014	0.571 ± 0.031	0.566 ± 0.034
MIL CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
Weighted CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
SC-MIL	0.619 ± 0.005	0.565 ± 0.025	0.559 ± 0.021

Table 10: PANDA native test results for validation-selected configurations, mean $\pm$ standard deviation across seeds. The two regimes carry different targets (binary detection vs. six-class ISUP), so their scores are not comparable.

4.5 Cross-Dataset Findings

Table 11 and Figure 4 collapse the four datasets to the comparison the report is built on, and three findings run across them.

Imbalance severity and mitigation headroom are decoupled—indeed inversely related here. CAMELYON16 is by far the most imbalanced dataset at patch level ($1 - H_{\text{norm}} \approx 0.832$, an order of magnitude above the others) yet shows among the smallest mitigation deltas ($+0.008$ macro F1 at patch level), because its detection task is near-ceiling. The largest positive deltas appear instead where imbalance is *mild* but the task is hard and classes are entangled: BRACS patch ($1 - H_{\text{norm}} \approx 0.072$, $\Delta = +0.039$) and PANDA WSI-bag ISUP grading ($1 - H_{\text{norm}} \approx 0.043$, $\Delta = +0.025$). Ranking these datasets by imbalance would predict the opposite of what is observed.

The patch/slide distinction is decisive, not incidental. CAMELYON16’s severe imbalance is entirely a within-slide, patch-level effect; at slide level its labels are nearly balanced ($1 - H_{\text{norm}} \approx 0.028$) and the slide-level detection task is exactly where every method ceilings. A large patch-level imbalance therefore creates no headroom once the representation separates the two classes, which is why disambiguating patch- from slide-level imbalance is necessary before reading any single ratio as a difficulty signal.

The same axis governs which method family wins and how well-calibrated it is. Where mild imbalance meets a hard, entangled patch task (TCGA-UT, BRACS), the methods that separate from the baseline are the ones that reshape the decision head or objective—CFAL prototypes

Dataset	Regime	Best mitigation method	Baseline macro F1	Δ macro F1
CAMELYON16	Patch	CFAL	0.958	+0.008
CAMELYON16	WSI bag	MDE-MIL / SC-MIL	0.965	+0.017
TCGA-UT	Patch	CFAL	0.870	+0.008
TCGA-UT	WSI bag	MDE-MIL	0.843	+0.014
BRACS	Patch	CFAL	0.451	+0.039
BRACS	WSI bag	MIL CE (baseline)	0.508	0.000
PANDA	Patch	CFAL	0.859	+0.007
PANDA	WSI bag	Balanced sampling	0.565	+0.025

Table 11: Cross-dataset mitigation headroom. Baseline is CE (patch) or MIL CE (WSI bag). Δ macro F1 is the best mitigation method’s mean test macro F1 minus the no-mitigation baseline; the BRACS WSI-bag row is 0.000 because no mitigation method beat MIL CE.

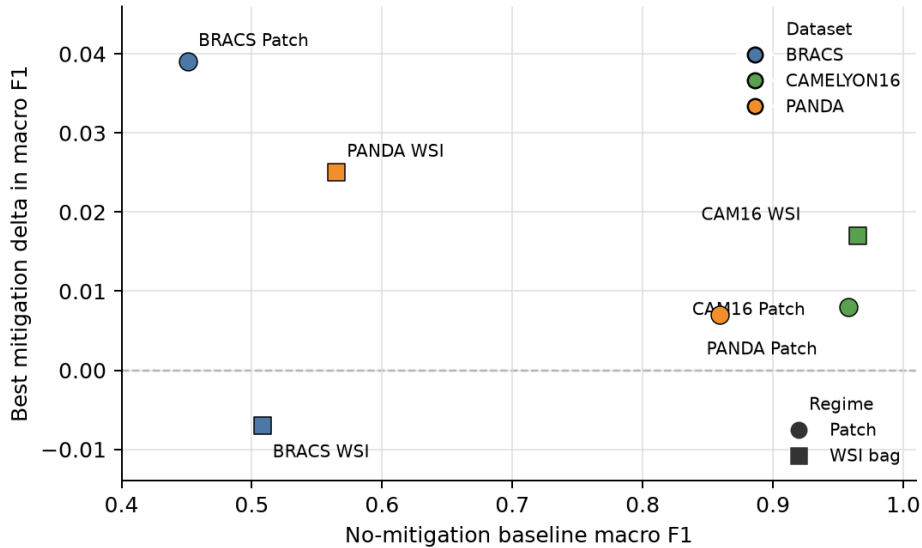


Figure 4: No-mitigation baseline macro F1 versus mitigation headroom across all dataset–regime pairs. High-baseline regimes leave little room for mitigation; lower-baseline regimes show the only material positive deltas, although BRACS WSI-bag shows that a low baseline alone does not guarantee an effective mitigation method.

and OKO set learning—while single-knob resampling, reweighting, and focal reweighting never separate from the baseline on any of the four datasets, and ProGAN augmentation lands within between-seed noise everywhere. Bag-aggregation methods (RankMix, MDE-MIL) help only where the bag task itself has headroom (TCGA-UT WSI). Calibration tracks the same separability axis: near-ceiling regimes reach post-scaling ECE near 0.001 (CAMELYON16 patch), while the hard grading and subtyping regimes retain residual post-scaling ECE up to 0.10–0.17, so a single property—class separability in the frozen representation—governs discrimination headroom and probability quality together.

5 Discussion

(a) **Frozen-encoder bottleneck.** Once a strong frozen pathology encoder is fixed, patch-level gains come from reshaping the decision head or objective rather than from resampling or reweighting. CFAL replaces the linear head with learnable class prototypes and a margin objective, pulling examples toward compact class centres and penalising confusion with nearby

wrong-class prototypes. OKO constructs training sets with one matched pair and k distractors; averaging per-example logits across set members implicitly regularises logit magnitude and suppresses the over-confidence that afflicts CE-trained classifiers, which explains its combination of high balanced accuracy and the best raw NLL and Brier among patch methods. The simpler reweighting and resampling variants remain within the CE overlap band, consistent with the interpretation that exposure corrections address only part of the error pattern when the representation already separates most classes. ProGAN augmentation does not deliver a clear advantage despite pixel-level FID comparable to the source study (Ruiz-Casado et al., 2024); the decisive difference is that the source method trains its classifier end-to-end on pixels, whereas this benchmark feeds synthetic patches through a frozen encoder that may map slightly off-manifold synthetic patches into embedding regions the head rarely sees for real data. On WSI bags, RankMix and MDE-MIL improve on plain MIL CE in every seed, because the bottleneck there includes instance selection and bag aggregation, which mixing informative subsequences (RankMix) and coupling natural and balanced branches with cross-expert consistency (MDE-MIL) can address even when the underlying features are strong.

(b) Separability, not imbalance, is the bottleneck. The four datasets show that the imbalance of the class distribution does not predict whether mitigation helps. BRACS is mildly imbalanced ($1 - H_{\text{norm}} \approx 0.072$ at patch level) but hard—fine-grained breast-lesion subtyping entangled in the frozen feature space—so only OKO and CFAL show a measurable patch-level lift. CAMELYON16 is severely imbalanced at patch level ($1 - H_{\text{norm}} \approx 0.832$) but easy—tumor and normal tiles are nearly separable in frozen Virchow2—so every method reaches near-ceiling discrimination and the unweighted baseline is competitive. PANDA provides the cleanest within-cohort control: two mildly imbalanced targets on the same biopsies diverge sharply in difficulty—binary detection (balanced accuracy ≈ 0.85) versus six-class ISUP grading (macro F1 ≈ 0.58)—while the no-mitigation baseline remains competitive in both. What governs the value of mitigation under a frozen-feature benchmark is thus class separability in the representation, not the imbalance of the distribution, and this is a direct caution against selecting or ranking imbalance-mitigation methods by imbalance ratio alone.

(c) ProGAN frozen-feature hypothesis. ProGAN augmentation is informative on BRACS and PANDA: the mild native imbalances never create the tail-scarcity regime where synthetic minority augmentation helped on TCGA-UT at harsher constructed severities. On BRACS (BAcc 0.460 versus CE 0.451) and PANDA (macro F1 0.855 versus CE 0.859), ProGAN lands within between-seed noise of the CE baseline, neither helping nor hurting. This is consistent with the frozen-feature explanation rather than a generator-quality failure: the generators reach reasonable pixel-level FID, but their outputs must pass through a frozen encoder that may map slightly off-manifold synthetic patches into embedding regions the head rarely sees for real data.

(d) Limitations. TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. Within each dataset the two benchmarks differ in prediction unit and per-slide evidence budget, so their result tables are non-transferable. The validation sweep varies one imbalance-specific control per method and does not exhaust architecture or training-schedule choices, and both regimes use frozen features rather than end-to-end encoder adaptation; the benchmark is controlled rather than a full reproduction of each original paper. Calibration results for the BRACS and PANDA WSI-bag regimes are additionally affected by small test sets and fine-grained label sets rather than method-specific weaknesses.

6 Conclusion

This report benchmarked representative class-imbalance mitigation methods on four histopathology datasets spanning the difficulty–imbalance plane, using frozen Virchow2 features, bounded per-slide evidence budgets, and patient-disjoint splits throughout, and characterising imbalance by the entropy-based $1 - H_{\text{norm}}$ at both patch and slide level. On CAMELYON16 every method reaches near-ceiling and the unweighted baseline is competitive despite the most severe patch imbalance in the suite; on TCGA-UT, CFAL reaches the highest patch-level macro F1 and OKO the highest balanced accuracy with the best raw NLL and Brier among patch methods, while RankMix and MDE-MIL lead the WSI-bag regime; on BRACS only OKO and CFAL separate from the CE baseline; and on PANDA both mildly imbalanced regimes are won by the no-mitigation baseline within between-seed noise. The practical implication is that the value of imbalance mitigation under a frozen-feature benchmark is governed by class separability in the representation, not by the imbalance of the class distribution. Imbalance mitigation should therefore be evaluated in the native input regime and alongside classwise recall and calibration, not only by aggregate discrimination, and method rankings should not be transferred across datasets on the basis of imbalance ratio alone.

A Additional TCGA-UT Analysis

This appendix collects the deeper TCGA-UT diagnostics referenced in Section 4.2: validation-selection provenance and tuning sensitivity, paired seed-level comparisons, classwise difficulty against support, and ProGAN augmentation diagnostics.

Regime	Method	Selected control	Val macro F1	Test macro F1	Test bal. acc.
Patch	CE	baseline (no sweep)	–	0.773	0.764
Patch	Balanced sampler	sampler_power=1	0.768	0.771	0.769
Patch	CE + soft F1 (balanced)	metric_loss_weight=256	0.781	0.780	0.781
Patch	CE + soft MCC (balanced)	metric_loss_weight=32	0.771	0.770	0.772
Patch	CFAL	cfal_sigma=1	0.792	0.798	0.789
Patch	Focal	focal_gamma=1.5	0.764	0.765	0.758
Patch	OKO	oko_k=4	0.783	0.788	0.791
Patch	ProGAN augmentation	final_depth_epochs=10	0.756	0.766	0.760
Patch	Weighted CE	weight_power=0.5	0.764	0.774	0.767
WSI bag	MIL CE	baseline (no sweep)	–	0.826	0.823
WSI bag	MDE-MIL (ensemble)	mde_mil_consistency_weight=2	0.851	0.865	0.849
WSI bag	Balanced MIL	sampler_power=0.75	0.821	0.831	0.829
WSI bag	Focal MIL	focal_gamma=0.5	0.823	0.824	0.823
WSI bag	Weighted MIL	weight_power=0.125	0.828	0.835	0.833
WSI bag	RankMix	rankmix_alpha=32	0.843	0.854	0.864
WSI bag	SC-MIL	sc_mil_temperature=0.1	0.828	0.839	0.837

Table 12: Validation-selected TCGA-UT configuration provenance. Configurations are chosen by mean validation macro F1 over seeds 0–2 with balanced accuracy as tie-breaker. A seed-blocked repeated-measures analysis of the validation sweeps shows across-configuration variation exceeding twice the seed standard deviation only for OKO (range-to-seed-noise ratio 75.8) and RankMix (ratio 2.06, $p < 0.01$); for all single-knob resampling, reweighting, and focal mechanisms the swept control produces variation within seed noise regardless of its value.

Comparison	Δ Macro F1	Δ Balanced accuracy
CE + soft MCC – CE	-0.003 ± 0.008 [+0.005, –0.000, –0.014]	$+0.009 \pm 0.007$ [+0.015, +0.012, –0.000]
CE + soft F1 – CE	$+0.007 \pm 0.010$ [+0.014, +0.014, –0.007]	$+0.018 \pm 0.008$ [+0.023, +0.023, +0.007]
OKO – CE	$+0.015 \pm 0.010$ [+0.024, +0.021, +0.001]	$+0.028 \pm 0.012$ [+0.038, +0.033, +0.011]
CFAL – CE	$+0.025 \pm 0.005$ [+0.030, +0.027, +0.019]	$+0.026 \pm 0.004$ [+0.030, +0.027, +0.020]
RankMix – MIL CE	$+0.028 \pm 0.010$ [+0.023, +0.041, +0.020]	$+0.041 \pm 0.010$ [+0.038, +0.055, +0.031]
MDE-MIL – MIL CE	$+0.039 \pm 0.016$ [+0.055, +0.043, +0.018]	$+0.026 \pm 0.016$ [+0.041, +0.034, +0.003]

Table 13: TCGA-UT paired validation-selected test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean $\pm$ SD across seeds; bracketed values list seed-ordered paired deltas. CFAL and OKO are above plain CE on macro F1 and balanced accuracy in every seed; on WSI bags, both RankMix and MDE-MIL are above plain MIL CE on both headline metrics in every seed. No formal p -values are computed.

B Native-Dataset Calibration Tables

Raw and temperature-scaled (TS) calibration for BRACS, CAMELYON16, and PANDA, referenced in Section 4. Each table gives ECE before and after scaling and the fitted temperature

Regime	Method	NLL	Brier	ECE
Patch	CFAL	2.850 ± 0.011	0.914 ± 0.001	0.787 ± 0.008
Patch	OKO	0.756 ± 0.040	0.270 ± 0.015	0.158 ± 0.013
Patch	CE + soft F1 (balanced)	3.038 ± 0.137	0.316 ± 0.008	0.155 ± 0.004
Patch	Weighted CE	1.965 ± 0.115	0.314 ± 0.016	0.147 ± 0.007
Patch	CE	1.976 ± 0.149	0.312 ± 0.018	0.147 ± 0.009
Patch	Balanced sampler	1.967 ± 0.139	0.313 ± 0.019	0.147 ± 0.009
Patch	CE + soft MCC (balanced)	2.870 ± 0.153	0.332 ± 0.017	0.162 ± 0.008
Patch	ProGAN augmentation	2.511 ± 0.093	0.327 ± 0.007	0.157 ± 0.004
Patch	Focal	1.370 ± 0.081	0.298 ± 0.014	0.131 ± 0.006
WSI bag	MDE-MIL (ensemble)	0.487 ± 0.005	0.166 ± 0.002	0.036 ± 0.004
WSI bag	RankMix	0.584 ± 0.029	0.170 ± 0.006	0.049 ± 0.001
WSI bag	SC-MIL	1.024 ± 0.145	0.204 ± 0.020	0.092 ± 0.009
WSI bag	Weighted MIL	1.224 ± 0.144	0.212 ± 0.012	0.099 ± 0.006
WSI bag	Balanced MIL	1.264 ± 0.165	0.223 ± 0.010	0.102 ± 0.005
WSI bag	MIL CE	1.283 ± 0.080	0.227 ± 0.024	0.106 ± 0.012
WSI bag	Focal MIL	1.117 ± 0.153	0.221 ± 0.014	0.098 ± 0.009

Table 14: TCGA-UT held-out test raw calibration metrics for validation-selected configurations over three seeds. Lower is better for NLL, Brier, and ECE. Values are not comparable across regimes. Table 15 gives the post-hoc counterpart.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	0.532 ± 0.029	0.220 ± 0.011	0.019 ± 0.003
Patch	OKO	0.617 ± 0.032	0.236 ± 0.011	0.029 ± 0.006
Patch	CE + soft F1 (balanced)	0.780 ± 0.026	0.263 ± 0.008	0.061 ± 0.028
Patch	Weighted CE	0.654 ± 0.030	0.252 ± 0.012	0.038 ± 0.005
Patch	CE	0.649 ± 0.035	0.250 ± 0.014	0.041 ± 0.002
Patch	Balanced sampler	0.653 ± 0.034	0.252 ± 0.013	0.038 ± 0.008
Patch	CE + soft MCC (balanced)	0.768 ± 0.034	0.271 ± 0.013	0.076 ± 0.029
Patch	ProGAN augmentation	0.723 ± 0.017	0.263 ± 0.005	0.073 ± 0.007
Patch	Focal	0.618 ± 0.025	0.251 ± 0.009	0.006 ± 0.004
WSI bag	MDE-MIL (ensemble)	0.471 ± 0.009	0.163 ± 0.003	0.023 ± 0.002
WSI bag	RankMix	0.428 ± 0.012	0.166 ± 0.005	0.028 ± 0.002
WSI bag	SC-MIL	0.440 ± 0.045	0.173 ± 0.018	0.007 ± 0.004
WSI bag	Weighted MIL	0.455 ± 0.033	0.177 ± 0.011	0.013 ± 0.004
WSI bag	Balanced MIL	0.479 ± 0.031	0.185 ± 0.010	0.013 ± 0.003
WSI bag	MIL CE	0.473 ± 0.018	0.188 ± 0.015	0.017 ± 0.010
WSI bag	Focal MIL	0.475 ± 0.024	0.189 ± 0.008	0.012 ± 0.002

Table 15: TCGA-UT held-out test calibration metrics after method-agnostic post-hoc temperature scaling. For each method and seed the temperature is fit only on that seed’s validation split and applied to the test split. Table 14 gives the raw-score diagnostic.

Regime	Class	Slides	Support rank	Difficulty rank	Recall	F1
Patch	Rectum adenocarcinoma	63	5	1	0.103	0.144
Patch	Cholangiocarcinoma	30	2	2	0.319	0.409
Patch	Esophageal carcinoma	113	10	3	0.362	0.420
Patch	Uterine Carcinosarcoma	71	7	4	0.539	0.519
Patch	Mesothelioma	70	6	5	0.556	0.628
WSI-bag	Rectum adenocarcinoma	63	5	1	0.338	0.409
WSI-bag	Esophageal carcinoma	113	10	2	0.523	0.564
WSI-bag	Cholangiocarcinoma	30	2	3	0.528	0.544
WSI-bag	Uterine Carcinosarcoma	71	7	4	0.625	0.611
WSI-bag	Stomach adenocarcinoma	274	18	5	0.693	0.735

Table 16: Hardest TCGA-UT classes under each regime’s CE baseline, averaged over seeds 0–2 on the held-out test split. Support rank is computed from full TCGA-UT slide counts (rank 1 = lowest support); difficulty rank orders classes by baseline error rate, $1 - \text{recall}$. Rectum adenocarcinoma is the hardest class in both regimes despite being the fifth-lowest-support class, and the rarest class (diffuse large B-cell lymphoma) is not among the five hardest; the rank association between slide support and CE-baseline recall is positive but modest (Spearman $\rho = 0.334$ for patches, $\rho = 0.301$ for WSI bags).

Class	Real train	Generated	Inception FID	Virchow2 mean NN
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	600	16,240	345.6	50.68
Cholangiocarcinoma	660	16,180	307.6	51.79
Pheochromocytoma and Paraganglioma	900	15,940	382.0	54.06
Uveal Melanoma	1,160	15,680	338.4	48.88
Rectum adenocarcinoma	1,350	15,490	239.5	47.97
Mesothelioma	1,430	15,410	286.3	51.23
Uterine Carcinosarcoma	1,550	15,290	304.0	50.94
Kidney Chromophobe	1,740	15,100	282.8	48.97
All augmented classes (31 total)	165,470	356,570	293.4 median	51.32 median

Table 17: ProGAN augmentation diagnostics for seed 0. “Generated” counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset; the class-specific generators reach FID 239.5–382.0 (median 293.4), comparable to the ProGAN FID reported by Ruiz-Casado et al. (2024). Virchow2 mean NN is the mean minimum L_2 distance from a fixed synthetic subsample to real same-class embeddings in the frozen feature cache.

ProGAN synthetic patches and nearest real training neighbors (seed 0, Virchow2 embedding space)

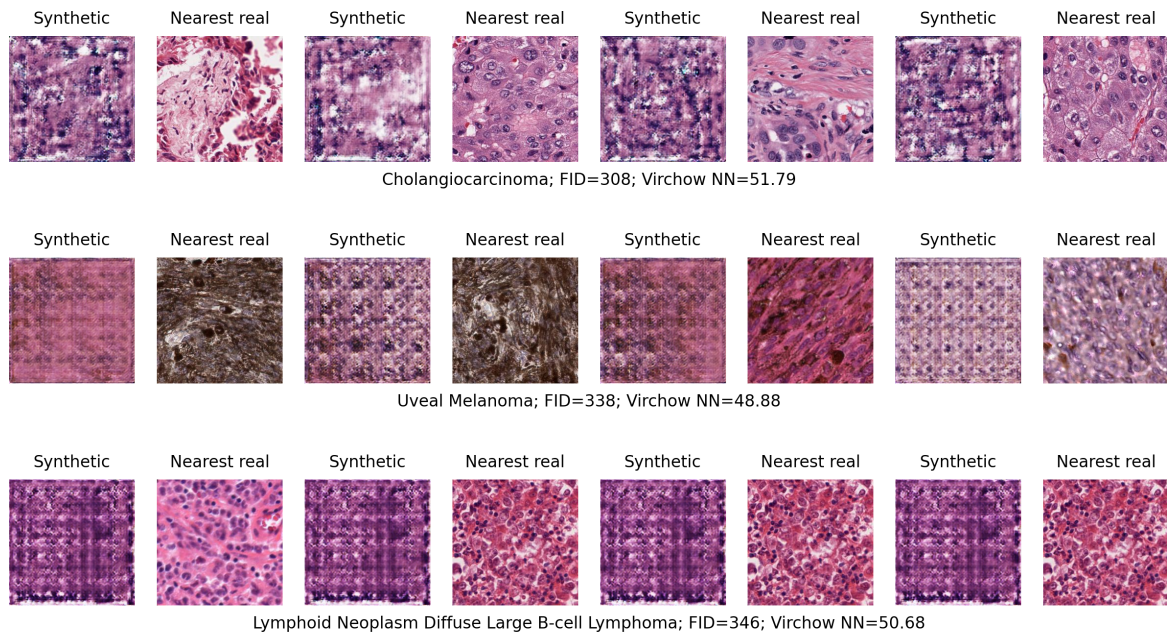


Figure 5: ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. The frozen-feature pipeline—not generator quality—is the proposed explanation for the null downstream result.

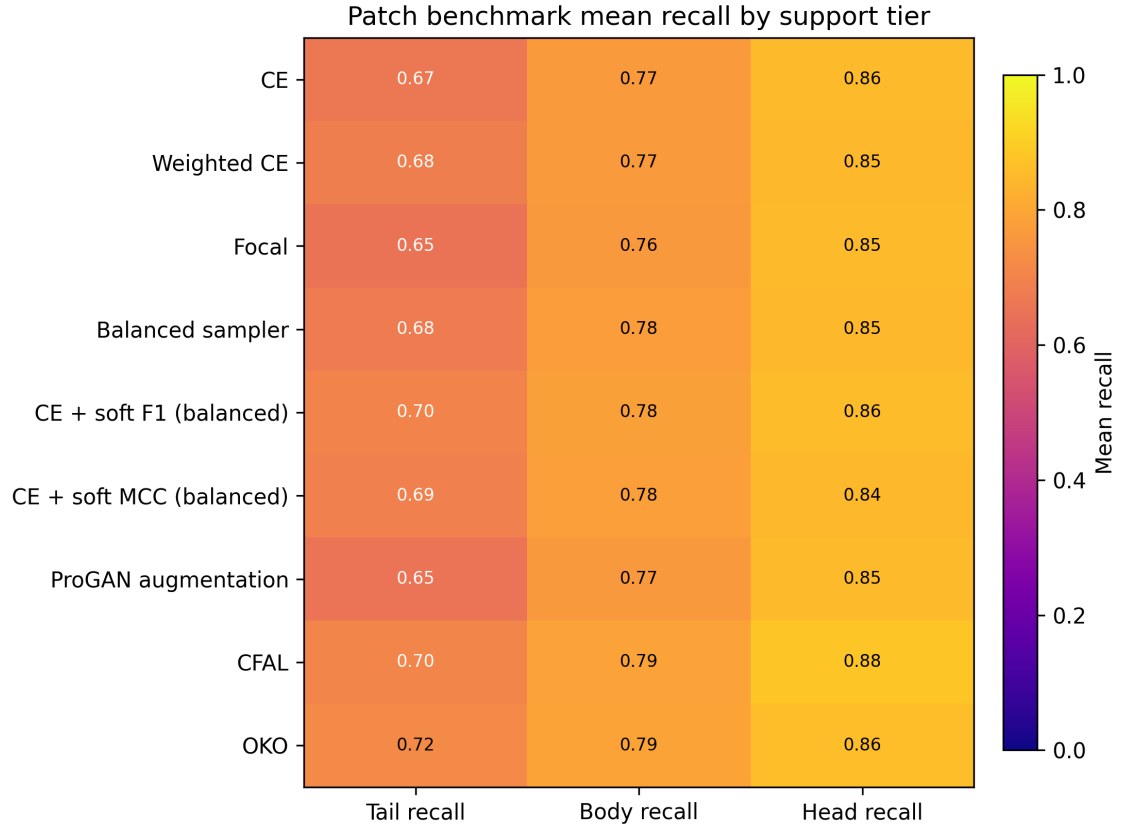


Figure 6: TCGA-UT patch-level mean recall by support tier on the held-out test split, exposing the head–body–tail variation hidden by aggregate accuracy.

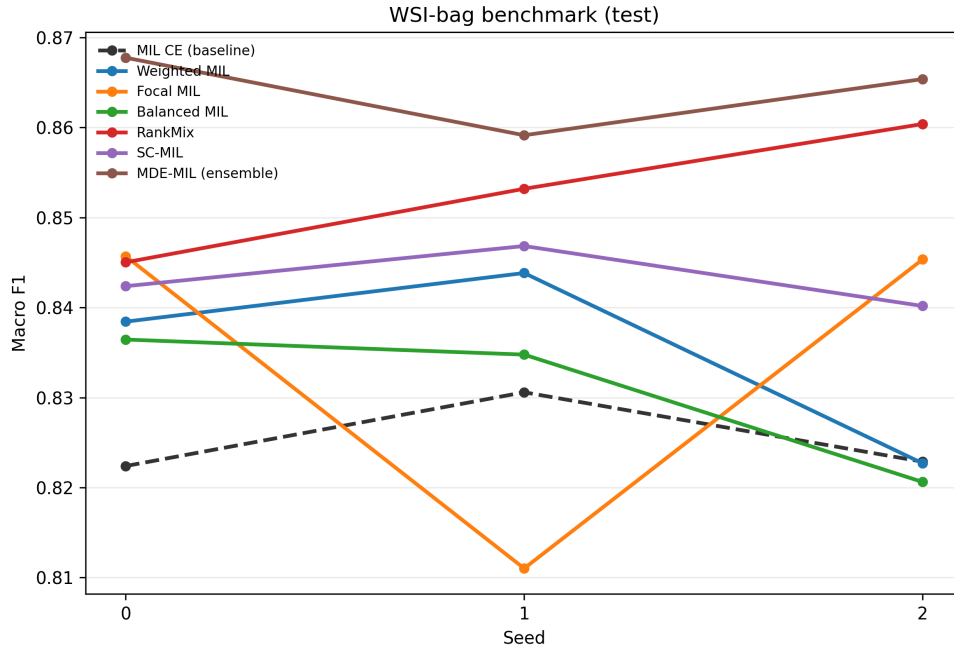


Figure 7: Validation-selected TCGA-UT WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE; the pronounced focal-MIL dip on seed 1 is a single-seed training instability.

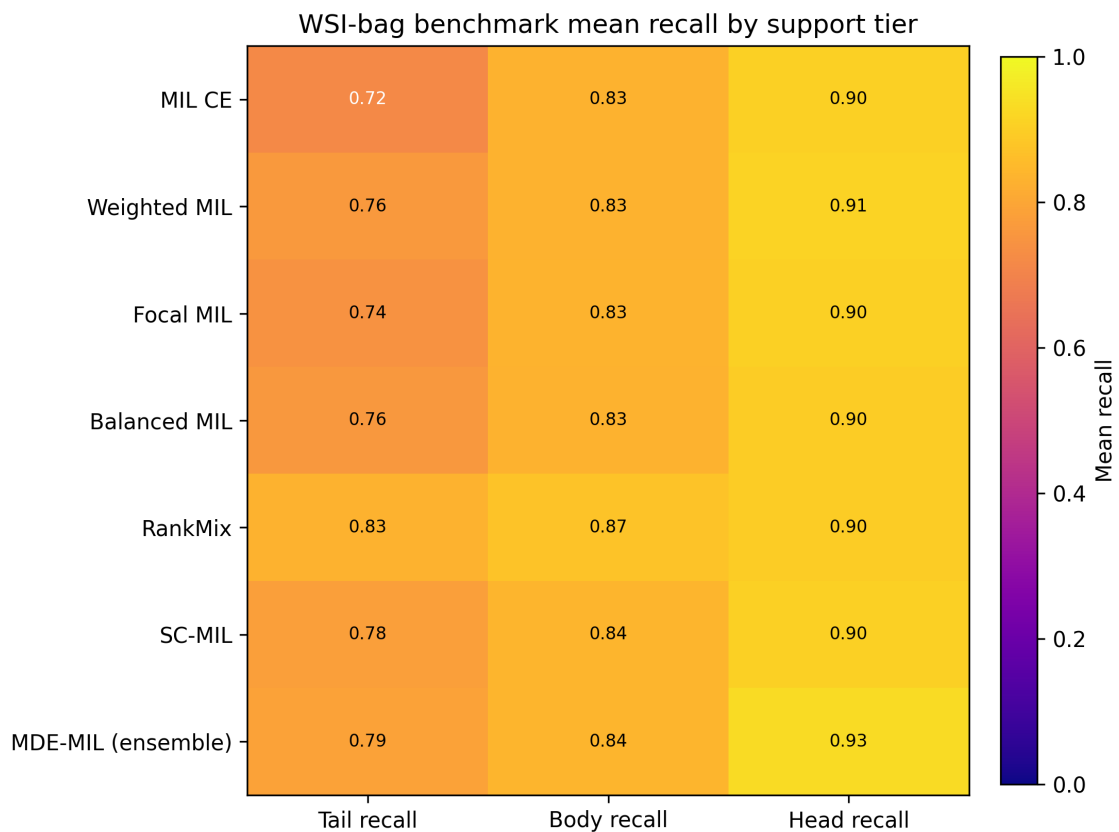


Figure 8: TCGA-UT WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient in the imbalanced setting.

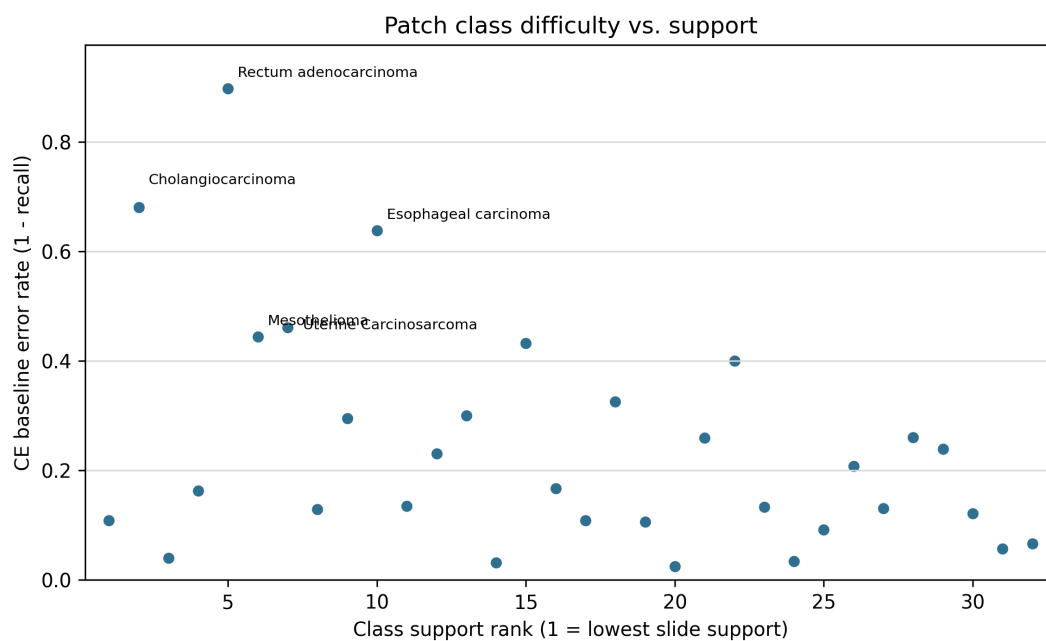


Figure 9: TCGA-UT patch CE-baseline class error rate versus full-dataset class-support rank. The hardest patch classes are not simply ordered by rarity.

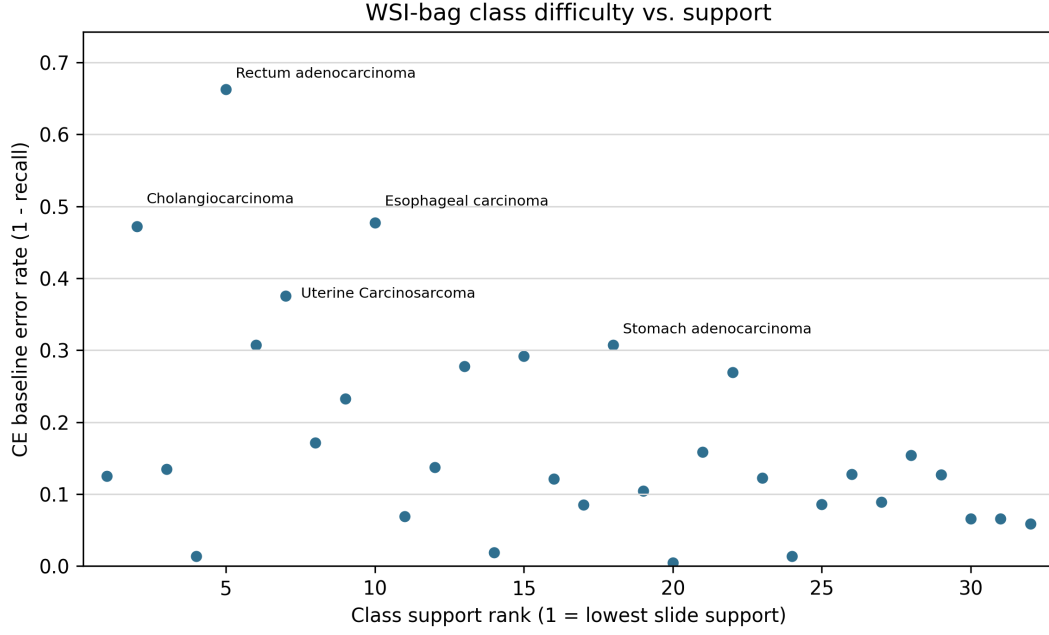


Figure 10: TCGA-UT WSI-bag CE-baseline class error rate versus full-dataset class-support rank. The same high-error classes are visible at slide level, showing that class difficulty is not reducible to class frequency.

T , mean $\pm$ standard deviation across seeds.

Method	ECE	ECE+TS	T
CE	0.354 ± 0.039	0.031 ± 0.010	5.812 ± 0.409
Weighted CE	0.367 ± 0.039	0.038 ± 0.017	5.752 ± 0.486
Focal loss	0.286 ± 0.023	0.030 ± 0.017	3.459 ± 0.405
CFAL	0.417 ± 0.026	0.049 ± 0.013	0.079 ± 0.010
CE + soft F1	0.405 ± 0.030	0.070 ± 0.027	8.532 ± 1.329
CE + soft MCC	0.366 ± 0.039	0.029 ± 0.011	6.582 ± 0.723
Balanced sampling	0.347 ± 0.036	0.021 ± 0.011	5.823 ± 0.601
OKO	0.045 ± 0.028	0.039 ± 0.014	0.877 ± 0.149
ProGAN augmentation	0.358 ± 0.041	0.039 ± 0.018	6.158 ± 0.677

Table 18: BRACS native patch-feature calibration.

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.012	0.165 ± 0.040	3.092 ± 0.322
Weighted CE	0.247 ± 0.051	0.163 ± 0.045	2.928 ± 0.343
Focal loss	0.164 ± 0.066	0.136 ± 0.010	1.977 ± 0.075
Balanced sampling	0.257 ± 0.064	0.168 ± 0.052	2.861 ± 0.314
RankMix	0.151 ± 0.074	0.144 ± 0.017	1.298 ± 0.108
SC-MIL	0.302 ± 0.097	0.168 ± 0.014	2.421 ± 0.242
MDE-MIL	0.277 ± 0.017	0.151 ± 0.010	2.083 ± 0.326

Table 19: BRACS native WSI-bag calibration.

Method	ECE	ECE+TS	T
CE	0.003 ± 0.001	0.001 ± 0.001	3.723 ± 0.142
Weighted CE	0.004 ± 0.002	0.001 ± 0.001	4.224 ± 0.503
Focal loss	0.003 ± 0.001	0.001 ± 0.000	4.547 ± 0.173
CFAL	0.464 ± 0.001	0.068 ± 0.004	0.050 ± 0.000
CE + soft F1	0.004 ± 0.001	0.001 ± 0.001	4.598 ± 0.179
CE + soft MCC	0.005 ± 0.001	0.002 ± 0.000	5.025 ± 0.191
Balanced sampling	0.003 ± 0.001	0.001 ± 0.001	4.071 ± 0.239
OKO	0.003 ± 0.000	0.001 ± 0.000	3.014 ± 0.100
ProGAN augmentation	0.003 ± 0.001	0.001 ± 0.001	4.119 ± 0.281

Table 20: CAMELYON16 native patch-feature calibration.

Method	ECE	ECE+TS	T
MIL CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Weighted CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Focal loss	0.021 ± 0.013	0.023 ± 0.010	0.050 ± 0.000
Balanced sampling	0.022 ± 0.009	0.023 ± 0.010	0.050 ± 0.000
RankMix	0.022 ± 0.030	0.028 ± 0.024	0.273 ± 0.386
SC-MIL	0.020 ± 0.018	0.017 ± 0.017	0.050 ± 0.000
MDE-MIL	0.013 ± 0.008	0.017 ± 0.017	0.050 ± 0.000

Table 21: CAMELYON16 native WSI-bag calibration.

Method	ECE	ECE+TS	T
CE	0.046 ± 0.007	0.035 ± 0.003	2.733 ± 0.236
Weighted CE	0.045 ± 0.005	0.036 ± 0.002	2.703 ± 0.225
Focal loss	0.024 ± 0.000	0.031 ± 0.002	1.404 ± 0.126
CFAL	0.382 ± 0.004	0.011 ± 0.003	0.060 ± 0.001
CE + soft F1	0.103 ± 0.009	0.009 ± 0.001	6.418 ± 0.324
CE + soft MCC	0.101 ± 0.008	0.008 ± 0.003	5.491 ± 0.183
Balanced sampling	0.072 ± 0.005	0.019 ± 0.003	3.642 ± 0.184
OKO	0.028 ± 0.008	0.005 ± 0.002	0.822 ± 0.027
ProGAN augmentation	0.042 ± 0.007	0.039 ± 0.002	2.595 ± 0.343

Table 22: PANDA native patch-feature calibration.

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Weighted CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Focal loss	0.129 ± 0.022	0.062 ± 0.020	2.002 ± 0.166
Balanced sampling	0.241 ± 0.021	0.039 ± 0.009	3.723 ± 0.142
RankMix	0.144 ± 0.009	0.045 ± 0.020	1.530 ± 0.029
SC-MIL	0.252 ± 0.020	0.070 ± 0.015	3.452 ± 0.297
MDE-MIL	0.239 ± 0.035	0.097 ± 0.039	2.113 ± 0.080

Table 23: PANDA native WSI-bag calibration.

C Terminology

Term	Meaning in this study
<i>Slide / WSI</i>	Diagnostic whole-slide image carrying one label.
<i>Case / patient</i>	TCGA participant identified by the first three fields of the TCGA slide barcode (e.g. TCGA-OR-A5J1); or the equivalent patient identifier in BRACS and PANDA.
<i>Patch</i>	256×256 H&E crop from a selected region; patch labels inherit the slide-level class.
<i>Patch embedding</i>	Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.
<i>WSI bag</i>	Set of patch-feature instances for one slide, consumed by the MIL model.
<i>Imbalance $1 - H_{\text{norm}}$</i>	Entropy-based class-imbalance measure in $[0, 1]$ (0 balanced, 1 maximally skewed); reported at patch and slide level (Queisler, 2026a).
<i>Head/body/tail class</i>	Fixed support tier from the full TCGA-UT slide distribution (8 each for head and tail; 16 body).
<i>Case-disjoint split</i>	Split rule preventing any participant’s slides, patches, or features from crossing train, validation, and test.
<i>Evidence budget</i>	Per-slide cap on the number of patch or bag instances; dataset-specific, following a common data-driven median rule (Table 1).

Table 24: Terminology used across all four dataset benchmarks.

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